CHAPTER 8

# Random variables and dispersion

**Chapter overview**

**Objectives.** Build a law; interpret events; calculate expectation, variance, deviation type ; use linearity ; simulate samples.
**Prerequisites.** Probabilities, weighted average, squares, trees and lists.
**Annual benchmark.** 19·28 January: 8 a.m.

## Diagnostic check: six questions, 10 minutes

Answer without consulting the course. Indicate the questions for which assistance is required. diagnostic check is not graded.

1. Calculate the average of 2, 4, 6.

2. Check if ( $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$) can be a law.

3. Calculate $(2 - 5)^2$.

4. If $P(A) = 0{,}6$, calculate $P(\overline{A})$.

5. A gross gain of 8 units and a bet of 3 give what net gain?

6. Calculate $\sqrt{0{,}25}$.

**After the diagnostic check.** Retrieve the concepts indicated with the teacher; repeat a question and then explain the rule used.

**Speaking and reasoning.** Explain why a theoretical average can be an impossible gain. the difference between a law, a simulation and a statistical series.

## Search before formalizing

Create two sets with three equiprobable issues with expectation 0 but separate standard deviations.
their laws and explain the difference.

**Organization.** Individual research 5 min, confrontation 2 5 min, pooling 5 min.
Keep a trace of a test, even if not completed. A conjecture must then be justified.

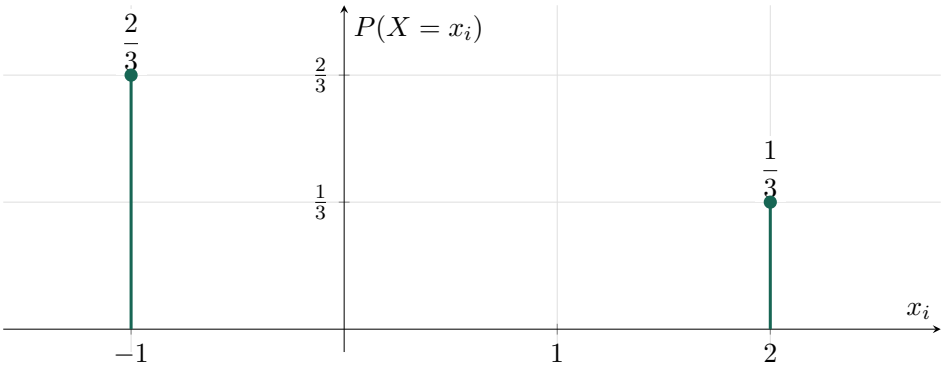
## 8.1 Variable, event and law

**Definition and results**

On a finite universe, a real random variable X is a function that combines a number real at every exit. $\{X = a\}$ collects the issues giving value a. Its law with each separate value $x_i$ probability $p_i = P(X = x_i)$, with $p_i \geq 0$ and $\sum p_i = 1$.

**Status and justification.** Definitions. Events associated with separate values form a partition of the universe; this justifies the sum of probabilities.

**Example solved.** A balanced dice gives an X gain equal to 2 for 1 or 2, and to $-1$ If not. The law is $P(X = 2) = \frac{1}{3}$, $P(X = -1) = \frac{2}{3}$. $\{X \leq 0\} = \{3; 4; 5; 6\}$ : Probability $2\frac{}{3}$.




The height indicates the probability: $P(X = -1) = \frac{2}{3}$ and $P(X = 2) = \frac{1}{3}$. Accurate probabilities are on both sticks.

**Point of vigilance**

Values are not always equiprobable even if the outcomes are.
which give the same gain.

**Immediate application.** Two balanced pieces are launched. X counts batteries. Write his law and $P(X \leq 1)$.

## 8.2 Expectation and linearity

**Definition and results**

Expectation is $E(X) = \sum_i x_i p_i$. It describes the theoretical long-term average, not a result
For variables X and Y on the same finite universe
and real a,b: $E(aX + b) = aE(X) + b$ and $E(X + Y) = E(X) + E(Y)$, no hypothesis independence.

**Status and justification.** Sleeping on the exits $\omega$, $E(X + Y) = \sum_\omega (X(\omega) + Y(\omega))P(\omega)$ ; distribute the sum gives the result. The constant b has expectation b because the probabilities have sum 1.

**Example solved.** For the previous law $P(X = 2) = \frac{1}{3}$, $P(X = -1) = \frac{2}{3}$, $E(X) = 0$ : the game is fair if X is the net gain. $Y = 3X - 2$, $E(Y) = 3 \times 0 - 2 = -2$.

**Point of vigilance**

A expectation is not an individual guarantee; a game can be lost on every horizon in spite of a positive expectation.

**Immediate application.** Gross gain equals 10 with probability $\frac{1}{5}$ and 0 otherwise. What bet makes the fair play?

## 8.3 Variance, standard deviation and König·Huygens

**Definition and results**

Noting $\mu = E(X)$, $V(X) = E((X - \mu)^2) = \sum_i (x_i - \mu)^2 p_i$. The standard deviation is $\sigma(X) = \sqrt{V(X)}$.
The formula of König·Huygens gives $V(X) = E(X^2) - E(X)^2$. A variance is always positive or zero; standard deviation has the same unit as X, variance a unit squared.

**Status and justification.** Develop ( $X - \mu)^2 = X^2 - 2\mu X + \mu^2$, apply linearity and $E(X) = \mu$ : $E(X^2) - 2\mu^2 + \mu^2$. We get the formula.

**Example solved.** For X values 0 and 4 with probabilities $\frac{3}{4}$ and $\frac{1}{4}$, $E(X) = 1$, $E(X^2) = 4$, $V(X) = 3$ and $\sigma = \sqrt{3}$. Direct calculation: $(0 - 1)^2(\frac{3}{4}) + (4 - 1)^2(\frac{1}{4}) = 3$.

**Point of vigilance**

$E(X^2)$ is not $E(X)^2$. A calculated negative variance indicates an error or a very small one error to be dealt with explicitly.

**Immediate application.** For X values $-1$, 1 equiprobable, calculate expectation, variance and Standard deviation.

## 8.4 Transform and Compare Gains

**Definition and results**

For real a,b, $V(aX+b) = a^2V(X)$ and $\sigma(aX+b) = |a|\sigma(X)$ A translation moves the average without changing the dispersion; a multiplication changes the scale. Two sets of the same expectation can have very different dispersions.

**Status and justification.** $aX+b-E(aX+b) = a(X-E(X))$. Raising squared and then taking expectation, we get $a^2V(X)$ The square root of $a^2$ is $|a|$.

**Example solved.** A game A always returns 2. A game B earns 0 or 4 equibably.
Two have expectation 2; variances 0 and 4. Their distributions are therefore not interchangeable despite the same average.

**Point of vigilance**

The variance of a sum is not, in general, the sum of the variances; no rule of this type n is used without assumptions.

**Immediate application.** $E(X) = 3$, $V(X)$ = 4. Give the characteristics of $Y = -2X + 5$.

## 8.5 Samples and mean observed

**Definition and results**

A sample size n is here a list of independent achievements of the same variable X. The mean observed m varies from sample to sample. $\mu = E(X)$ ; of repeats make it possible to observe the proportion of satisfactory samples $|m-\mu| \leq \frac{2\sigma}{\sqrt{n}}$. This threshold is an experimental task of the programme, not an assertion of coverage rates Right.

**Status and justification.** Experimentation: here no convergence theorem or law is shown
The list size, the calculation of the average and the distinction between n and the number are checked. No samples.

**Example solved.** For X of 0 and 2 equiprobable values, $\mu = 1$, $\sigma$ = 1. For n=100, threshold $\frac{2}{10} = 0{,}2$ : averages are counted in [0 ,8; 1,A result of 956 out of 1000 means frequency 0.956 for this simulation only.

**Point of vigilance**

Do not copy a single draw once; do not announce that the proportion is necessarily valid 95 %.

**Immediate application.** For $\mu = 5$, $\sigma$ = 3 and n=36, give the experimental interval requested.

## Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

### Fiscal year 8.1 · A law to be built

A balanced dice with six faces numbered from 1 to 6 is launched. $X$ associated with the result
the rest of its Euclidean division by 3. For example, a result divisible by 3 gives $X = 0$.

1. For each of the six sides, indicate the value taken by $X$.

2. Presenting the distribution of probability of $X$ in a table and verify that the sum of probabilities equals 1.

3. Calculate $P(X < 2)$ specifying which values of $X$ are retained.

### Exercise 8.2 · Two variance calculations

Random variable $X$ takes the values $-2$, 0 and 4 with respective probabilities $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{1}{4}$.

1. Calculating His Expectation $\mu = E(X)$ and the moment $E(X^2)$.

2. Calculate $V(X)$ using the König-Huygens formula, then deduct the exact standard deviation.

3. Find variance from definition $V(X) = E((X-\mu)^2$Presentation of the three contributions the calculation.

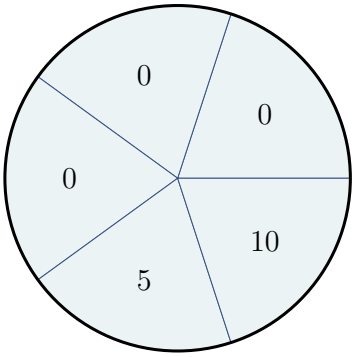
### Exercise 8.3 · Fair play

A wheel has five sectors of the same angle, bearing numbers 0, 0, 0, 5 and 10.
the five equiprobable sectors. $G$, in monetary units.
To play, we pay a bet. $m$ ; the net gain is therefore $X = G - m$.

1. To establish the law of $G$, then determine the bet $m$ which makes the game fair, i.e. such that $E(X) = 0$.

2. After studying the variance, calculate the standard deviation of the gross gain. where the bet is fair.

3. Explain why changing the bet changes the expectation of the net gain but not its standard deviation.




The five sectors have the same angle. The three sectors bearing 0 are distinct and all must be Counted.

### Exercise 8.4 · Same average, other risk

Two fictitious investments are proposed. The first one provides with certainty a gain $X = 1$. The second provides a gain $Y = -2$ with probability $\frac{1}{2}$ and one gain $Y = 4$ with probability $\frac{1}{2}$. The winnings are expressed in the same currency unit.

1. Compare the expectations of the two gains, then their variances.

2. One person states: "These investments are equivalent since they have the same average." Explain what this sentence omits and what the variance allows here to distinguish.

### Exercise 8.5 · Linearity without independence

We are launching a balanced dice numbered from 1 to 6. $X$ the result of the launch and it defines $Y = 7 - X$. The two variables are thus obtained from the same launch.

1. Calculate $E(X)$ and $E(Y)$, then $E(X + Y$ Check the linearity of expectation on this example.

2. Variables $X$ and $Y$ Are they independent? Justify using, for example, events $\{X = 1\}$ and $\{Y = 6\}$.

3. Does the linearity of expectation require independence? Conclude based on your results.

### Exercise 8.6 · Minimum loss

A Random Variable $X$ takes a finite number of values and has a expectation $\mu$. For a real $a$, mean quadratic error associated with constant prediction is measured $a$ by $E((X - a)^2)$.

1. Demonstrate that $E((X - a)^2) = V(X) + (a - \mu)^2$. We can write. $X - a = (X - \mu) + (\mu - a)$.

2. Deduct the value of $a$ which minimizes this error, then the minimum value. Justify the uniqueness of the optimal choice.

### Exercise 8.7 · Frequency of Letters

We want to measure the frequency of the letter "a" in a text. The denominator is the number of Alphabetical characters: spaces, numbers and punctuation signs are excluded.
are assimilated to the tiny ones; the accented letters remain distinct from "a".

1. Write a Python function that returns this frequency. Specify the selected result when the text contains no letters.
2. Test function with `Aba !`, then with `123 !` and with the empty string. Explain the results before running the programme.

### Fiscal year 8.8 · Deferred review

There are two independent repeats of an experience whose probability of success is worth $\frac{1}{2}$. The random variable $X$ counts the number of successes achieved.

1. Retrieving the law of $X$ using the four possible results. Do not use a formula general distribution of binomial distribution.
2. Calculate $E(X)$ and $V(X)$ on the basis of this law.

**Personal work.** Perform only the task assigned at the end of the meeting, in progress Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions Resuming exercise 8 a week after the chapter (10 min). is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

## To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

**CQFD, Exercise 28 p. 306, all questions.** Calculate expectation from the table of law; maintain the probabilities associated with each value.

**CQFD, exercise 31 p. 306, all questions; use the law of fiscal year 28.** Calculate the then the standard deviation, respecting the rounding to the tenth requested by the manual.

## MCQ 1: Understand and Choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. A distribution of probability has a sum of probabilities equal to

   A. $0$
   B. $1$
   C. $n$
   D. the average

2. X is 0 and 4 with probabilities $\frac{3}{4}$ and $\frac{1}{4}$; $E(X) =$

   A. $1$
   B. $2$
   C. $3$
   D. $4$

3. $V(X)$ is

   A. $E(X^2)$
   B. $E(X)^2$
   C. $E(X^2) + E(X)^2$
   D. $E(X^2) - E(X)^2$

4. If $V(X) = 9$, $\sigma(X) =$

   A. $81$
   B. $9$
   C. $3$
   D. $\frac{1}{3}$

5. $E(2X + 1)$ is

   A. $2E(X)$
   B. $2E(X) + 1$
   C. $E(X) + 1$
   D. $E(X)^2 + 1$

## MCQ 2: understand and choose

Five teaching questions with one answer. Choose one answer per question; then explain one
This chapter format does not reproduce the official format of the EAM.

1. $V(X + 5)$ is

   A. $V(X) + 5$
   B. $V(X) + 25$
   C. $V(X)$
   D. $5V(X)$

2. $\sigma(-3X)$ is

   A. $3\sigma(X)$
   B. $-3\sigma(X)$

C. $9\sigma(X)$

D. $\sigma(X)$

3. Expectation is necessarily

A. a possible value

B. One integer

C. Positive

D. None of these three statements

4. $E(X+Y)=E(X)+E(Y)$ requires

A. Independence

B. No independence

C. X=Y

D. Positive X

5. For $\sigma = 2$, n=100, $\dfrac{2\sigma}{\sqrt{n}} =$

A. $4$

B. $0{,}04$

C. $0{,}4$

D. $2$

## True or false?

1. A variance may be negative.

2. Two variables of the same expectation have the same law.

3. Adding a constant to a variable does not change its standard deviation.

**Personal review**

I construct a complete law; I distinguish crude and clear; I control the two formulas of variance; I do not Confound not frequency and probability.
For each objective, indicate: acquired without help; succeeded with help; to rework. Write an example and a mistake to avoid.